IMPERIAL COLLEGE LONDON

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EE9-CS2-1
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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Tuesday, 5 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	D. Angeli
	Second Marker(s) :	E.C. Kerrigan

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1. Consider the following two-dimensional time-invariant nonlinear autonomous system:

$$\begin{aligned} \dot{x}_1 &= -x_1^3 + 2x_1 + x_2 \\ \dot{x}_2 &= -x_2^3 + 2x_2 + x_1 \end{aligned}$$

with state $x = [x_1, x_2]'$ evolving in $\mathbb{R}^2$.

a) Discuss existence and uniqueness of solutions. [2]

b) Find and sketch the nullclines of the vector-field (Hint: for a better result find intersection with axes and slope when crossing 0). [3]

c) Find regions in $\mathbb{R}^2$ delimited by nullclines and identify the orientation of the vector-field in each one of them (Hint: there should be 8 of them). [3]

d) Find, if any, forward invariant regions among those identified in part c). [2]

e) Find all equilibria of the system (Hint: exploit symmetry to find equilibria analytically). [2]

f) Linearize the system around each equilibrium and discuss the local phase-portrait (discuss hyperbolicity). [3]

g) Show that the diagonal $\{x \in \mathbb{R}^2 : x_1 = x_2\}$ is invariant for the system. [3]

h) Sketch the global phase portrait, emphasizing equilibria and how their local portraits merge into each other. [2]

2. Consider the real matrices given below

$$A_1 = \begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix} \qquad A_2 = \begin{bmatrix} -1 & 0 \\ 1 & -2 \end{bmatrix}.$$

a) Find a real symmetric and positive definite matrix $P \in \mathbb{R}^{2\times 2}$ such that the two Matrix inequalities given below are simultaneously fulfilled:

$$A_1'P + PA_1 < 0 \qquad \text{and} \qquad A_2'P + PA_2 < 0.$$

[5]

b) Consider the bidimensional nonlinear system Σ_1:

$$\Sigma_1 : \qquad \dot{x} = dA_1x + (1-d)A_2x,$$

with input $d \in \mathbb{R}$ and state $x \in \mathbb{R}^2$. Show that it admits an equilibrium in 0 regardless of the value of d (i.e. d enters multiplicatively). [1]

c) Prove, by finding a suitable quadratic Lyapunov function, that 0 is a Uniformly Globally Asymptotically Stable equilibrium with respect to disturbances $d \in [0,1]$. [4]

d) Consider next the scalar nonlinear system Σ_2:

$$\Sigma_2 : \qquad \dot{z} = -4z^3 + z, \qquad y = (1+z)/2$$

with state $z \in \mathbb{R}$ and output y. Find, for all initial conditions z_0 in $\mathbb{R}$, the corresponding ω-limit set $\omega(z_0)$. [4]

e) Define the cascade interconnection between Σ_2 and Σ_1 (viz. let $d = y$; in words the output of the Σ_2 subsystem acts as an input to Σ_1), and consider the associated state-space representation with state $[x', z]' \in \mathbb{R}^3$. [2]

f) Find, for the cascaded system with an arbitrary initial condition $[x', z]' \in \mathbb{R}^3$, the correspoding ω-limit set (justify your answer). [5]

3. Consider the two-dimensional time-invariant system specified below:

$$\begin{aligned} \dot{x}_1 &= 4x_2 \\ \dot{x}_2 &= -2x_1 - 2\sin(x_1)\cos(x_1) + u, \\ \\ y &= 4x_2. \end{aligned}$$

a) Show that the system is lossless and passive for a suitable choice of storage function $S(x)$ (Hint: notice that the system is affine). [5]

b) Design a static output feedback $u = k(y)$ that achieves global asymptotic stability of the origin, while fulfilling a constraint on the input signal amplitude: $|u(t)| \leq 1$, for all t. (*Hint: exploit previous storage function as a candidate Lyapunov function and use a passive feedback*). [5]

c) Choose next a static state-feedback, $u = \tilde{k}(x)$ capable of steering the system's solution towards a prescribed level of energy S_{ref}. (Hint: consider the equation for $\dot{S}$ as a scalar system to be controlled. Use as a candidate Lyapunov function the quantity $W(x) = (S(x) - S_{ref})^2$ and choose u so as to get a derivative that is negative semidefinite). [5]

d) Argue that the feedback considered yields asymptotic convergence of solutions towards the desired level of energy S_{ref}, namely

$$\lim_{t \to +\infty} S(x_{cl}(t)) = S_{ref}$$

where $x_{cl}(t)$ denotes the solution of the closed-loop system. (Hint: assume $x(0) \neq 0$ and exploit monotonicity of the Lyapunov function along solutions). [5]

4. Consider the following nonlinear SISO control system:

$$\begin{aligned}\dot{x}_1 &= (x_1 - x_2)/2 + \text{atan}(x_1) + (2 + \cos(x_2))u \\ \dot{x}_2 &= (x_1 - x_2)/2 - \text{atan}(x_1) - (2 + \cos(x_2))u \\ \dot{x}_3 &= -x_3^3 + x_1 + x_2 \\ \\ y &= x_1 + x_2,\end{aligned}$$

with state $x \in \mathbb{R}^3$, input u and output y.

a) Compute the relative degree of the system and explain whether this is locally or globally defined. [4]

b) Find an I-O linearizing feedback law, of the form $u = k(x, v)$ where v is the new auxiliary input. [3]

c) Bring the system in normal form and highlight the internal dynamics. [5]

d) Discuss the Input-to-State Stability of the internal dynamics. [4]

e) Choose a linear stabilizing output feedback for the I-O dynamics and discuss the stability of the resulting cascaded system in the light of stability results on interconnected systems. [4]